Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director picks the next game's two tags. Had no pair passed the selection rule, they would pick a crowded one (table, last row).

2 Why a network

Counts can't show which tags sit in separate parts of the tag network, or which tag links them. The network can, through shared neighbours and bridges.

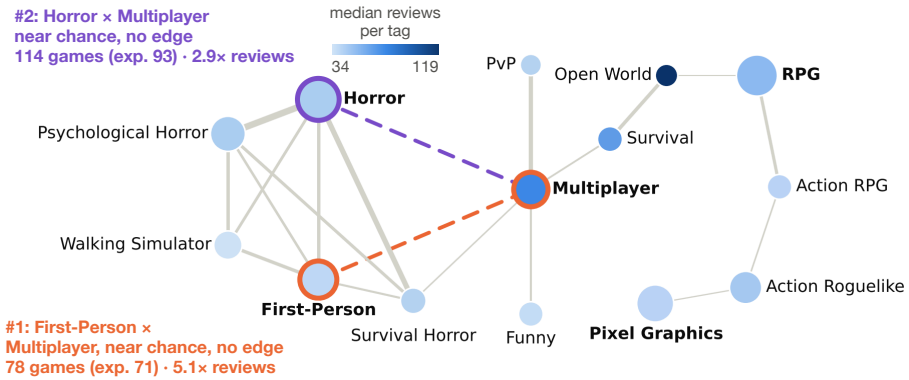
3 Data

Source	S1 FronkonGames steam-games-dataset (Hugging Face); S2 live Steam store.
Obtained	S1 download (Sep 2025). S2: 1,000 random S1 games (seed 7202), scraped 2026-10-14.
Legal	S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s; no personal data.
Volume	S1: all 9,094 Indie games released 2023–24 with ≥10 reviews. S2: 996 with a live page.
Check	S1 vs S2: mean top-5 tag Jaccard 0.87; Spearman ρ of reviews 0.98.
Bias	Favours games that sold; reviews = buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	Co-op, Online/Local Co-Op, Local Multiplayer → Multiplayer (84% of Horror × Multiplayer games were co-op too).
Weight	NPMI: co-occurrence vs chance (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0.
Window	Top 5 tags per game, as the store page shows; with all 20, #1's reviews drop from 5.1× to 3.0×.
Excluded	"Indie"; Early Access, Free to Play, adult tags; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**.

5 The picture



14 of 80 tags: ranked pairs (bold), #1 and #2's top-2 neighbours, linking paths. Size = games, width = NPMI. *Not shown.

Selection rule: of pairs on ≥73 games (0.8%, for stable medians), keep the **not crowded** (games ≤ 1.5× chance = games with A × games with B ÷ 9,094) with **separate neighbourhoods** (≤20% shared); rank by median reviews vs the median game (40).

Rank: pair	Games (chance)	Crowding	Shared neighbours	Median reviews	2025 hold-out
#1: **First-Person × Multiplayer**	78 (71)	1.10×	8%	204 (5.1×)	1.14× · 2.4×
#2: **Horror × Multiplayer**	114 (93)	1.23×	8%	117 (2.9×)	1.45× · 2.7×
#3: **Pixel Graphics × RPG**	133 (107)	1.25×	0%	70 (1.8×)	1.27× · 1.0×
Crowded*: **Building × Simulation**	74 (29)	2.54×	8%	136 (3.4×)	2.46× · 2.5×

6 Findings

- **#1** is near chance (NPMI 0.02, no edge) yet gets 5.1× the median reviews (permutation test $p < 0.001$); 68% are horror.
- **Survival Horror is the only bridge:** remove it and First-Person and Multiplayer go from 2 to 5 steps apart.
- **Holds up:** #1 at all 8 edge cutoffs; above the median in 100/100 bootstraps, but #1 in only 59; 3.4× without its top 5 games; passes on held-out 2025 releases (table: crowding · reviews).

7 Recommended action

- Week 0: pitch a first-person multiplayer horror game (#1).
- Weeks 1–6: prototype. Week 7: Coming Soon page tagged First-Person, Multiplayer, Survival Horror.
- Every quarter: re-run on new releases (section 8).

8 What would change my mind

#1 and #2 are one multiplayer-horror bet, and First-Person is spreading (27% of 2025 games). If #1 passes 1.5× crowding, I switch to #3 (Pixel Graphics × RPG; only 1.0× in 2025).